G V S S Krishna Mowli

ee25mtech11019@iith.ac.in | krishnamowli003@gmail.com | +91 9363138266 | LinkedIn | GitHub

EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.96
Krishnaswamy Vidyaniketan	Class XII (CBSE)	2021	92.60%
Krishnaswamy Vidyaniketan	Class X (CBSE)	2019	89.20%

SKILLS

Programming & Software: Python, MATLAB, ROS, Gazebo, C, Verilog, SQL (Basic), Git, GitHub, Linux, LaTeX

VLSI, FPGA & Semiconductor: Digital VLSI, Analog IC Design, CMOS, PDSOI, FPGA, RTL Design, Verilog, Cadence Virtuoso, Vivado, Sentaurus TCAD, Logic Synthesis (Basic).

Embedded Systems & IoT: Embedded Systems, Arduino, Raspberry Pi, IoT, Sensors, Actuators, ADC/DAC, PWM, UART, I²C , CAN.

Robotics, AI & Machine Learning: Multi-Agent Systems, Swarm Robotics, Reinforcement Learning, Federated Learning, PPO.

Control, Optimization & Systems: Control Theory, State-Space Modeling, PID Control, LQR, MPC, Optimal Control, Robust Control, Adaptive Control, Nonlinear Control, Distributed Control, State Estimation, Kalman Filter, Lyapunov Stability, System Identification, Convex Optimization, QP, SOCP, LMI, Dynamic Programming.

INTERNSHIPS

FPGA & DSP Design Research Intern | NIT Tiruchirappalli

- Designed and verified a **4-level pipelined architecture** and **Braun multiplier** for hardware implementation on a **Xilinx Artix-7 FPGA**, executing comprehensive synthesis, placement, and timing closure in **Vivado**.
- Developed modular **Verilog DSP processor cores**, integrating instruction-fetch logic, **program memory interfaces**, stack execution pipelines, and synchronous read/write control registers for memory subsystems.
- Constructed self-checking **RTL behavioral testbenches** and functional simulation suites in **Vivado**, validating cycle-accurate processor execution and eliminating internal race conditions before hardware sign-off.

IoT Systems Intern -- RINEX | IITR & IITBbh

- **Developed IoT prototypes** for environmental sensing & device control using **Arduino, Adafruit, and MIT App Inventor**.
- **Built an Arduino-based RC car** implementing **forward, reverse, left, right, and stop** controls.
- **Developed a Telegram-integrated IoT chatbot** using **Python and Adafruit IO** for remote light control; explored **Speech-to-Text and ThingSpeak** for voice control and data visualization.

Image Processing & CNN Internship | NIT Tiruchirapalli

- Explored image classification using **Python, PyTorch, FER2013, CK+, VGG19, and ResNet18**.
- Implemented computer vision workflows for preprocessing, augmentation, training, validation, and evaluation.
- Engineered HOG, LBP, and colour-histogram features and evaluated **SVM and Random Forest** classifiers.
- Analyzed image classification through **feature engineering, CNN architecture, and model evaluation**.

PROJECTS

Fault-Tolerant Turbo Decoder for SRAM-Based FPGAs

Verilog -- FPGA -- Digital VLSI -- Fault Tolerance -- Vivado

- Designed and analyzed a fault-tolerant Turbo Decoder architecture for SRAM-based FPGAs capable of mitigating

Single Event Upsets (SEUs) in both configuration and user memories.
- Implemented a Duplication-With-Comparison (DWC) architecture that detects decoder inconsistencies using decoded outputs, Log-Likelihood Ratio (LLR) confidence metrics, and sign-change monitoring.
- Evaluated architectural trade-offs between DWC and Triple Modular Redundancy (TMR), demon strating significantly lower hardware overhead while maintaining high fault tolerance.
- Investigated FPGA fault-injection methodologies and reliability-aware digital design techniques for radiation-prone aerospace and satellite communication systems.

Ultra-Low-Power MPPT Architecture for Battery-less IoT Energy Harvesting

Cadence Virtuoso -- Analog IC Design -- Energy Harvesting

- Architected an **ultra-low-power Time-to-Voltage Boost Converter** in **Cadence Virtuoso** using **UMC 45nm CMOS** to maximize photovoltaic energy harvesting across battery-less wireless IoT sensor nodes.
- Designed an **adaptive step-size Hill-Climbing controller** with **VCO-based perturbation logic**, substantially accelerating dynamic tracking speed while suppressing parasitic steady-state power oscillations.
- Optimized front-end **quiescent current dissipation** and verified layout parasitics, achieving a verified **99.9985% MPPT tracking efficiency** across broad dynamic irradiance operating ranges.

Tax-Aware Portfolio Optimization under Tax and Transaction-Cost & E.S.G. Constraints Portfolio Optimization

SOCP/QP -- Python -- Financial Engineering

- Designed a comprehensive portfolio optimization framework integrating capital gains taxation, trans action costs, wash-sale constraints, and ESG objectives into a unified convex optimization model.
- Formulated the investment problem as a sequence of Second-Order Cone Programming (SOCP) and Quadratic Programming (QP) problems to efficiently compute globally optimal portfolios.
- Implemented a two-stage optimization strategy for tax-efficient rebalancing, multi-account asset lo cation, and sign-constrained refinement while maintaining portfolio diversification.
- Evaluated the framework across multiple investment scenarios, demonstrating effective balancing of risk, return, taxation, trading costs, and sustainability objectives.

Distributed Consensus in Multi-Robot Systems with Collision & Obstacle Avoidance - Ongoing Thesis

- **Developing a decentralized multi-robot consensus framework** using **weighted graph theory and Laplacian-based coordination** for position agreement and distributed field-parameter estimation.
- **Formulating probabilistic collision-avoidance guarantees** using **Control Barrier Functions and Cantelli's inequality**, deriving pointwise and mission-horizon bounds with a **fixed safety margin**.
- **Implementing gradient-descent field estimation with basis functions** and validating the proposed safety guarantees through **Monte Carlo simulations** against empirical collision probabilities.

POSITIONS OF RESPOSIBILITY

Teaching Assistant - Control Systems | IIT Hyderabad

- **Guided students** in Control Systems through course assignments and academic activities, while supporting **examinations, invigilation, and evaluation** under the faculty instructor.

Placement Coordinator - Systems & Control | IIT Hyderabad

- **Coordinated placement activities** for the Systems & Control specialization, serving as the primary liaison among **recruiters, students, faculty, and the Institute Placement Office**.

Volunteer - Prayas, Rural Development Center (RDC) | IIT Hyderabad

- Volunteering with the Rural Development Center (RDC) under Prayas, with upcoming involvement in **planning and organizing community-development initiatives** for rural and nearby communities.

Publicity Coordinator - Probe'23 | NIT Tiruchirappalli

- **Led outreach to 50+ colleges** through virtual and on-site campaigns for the ECE department's national-level symposium, contributing to **1,500+ student footfall.**

RELEVANT COURSES

Multi-Agent Systems | Reinforcement Learning | Machine Learning & Applications | Fundamentals of Robotics

Network Analysis & Synthesis | Digital Circuits & Systems | Electronic Circuits | Analog Integrated Circuits | VLSI

Systems

VLSI Design: RTL to GDS | Embedded Systems | Microprocessors & Microcontrollers | Semiconductor Physics & Devices

Optimal Control | State Estimation | Introduction to State-Space | Convex Optimization | Advanced Control Lab |

Matrix Theory | Random Variables & Stochastic Processes | Advanced Stochastic Processes

HOBBIES AND INTERESTS

Financial Markets: Self-directed study and research on stock markets and investing

Sports : Table Tennis, Cricket, Chess, Badminton, Basketball, Volleyball, Carroms

Astrology: Independent study and research of astrology and horoscopes of several individuals

Reading: Diverse genres spanning fiction, philosophy, self-development, history, and Indian literature

CERTIFICATIONS

Semiconductor Specialization

1. Semiconductor Physics
2. Diode- PN Junction and Metal Semiconductor Contact
3. Transistor- Field Effect Transistor and Bipolar Junction Transistor

Matrix Algebra

Introduction to Basics Vibrations